Auralis Tech

Group Components

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Project Overview

Name of the shop: Auralis Tech

Project Rationale & Sourcing

For Auralis Tech, our core business focuses on the distribution of both hardware and software. After evaluating several industry-leading platforms, we established the following criteria for our product sourcing:

- **PCComponentes:** Although recognized for its extensive catalog of laptops and licenses, its massive inventory was deemed beyond the current scope of this project.

- **CoolMod (Primary Source):** Selected as our main target for web scraping. Its interface is clean and well-structured, providing a curated selection of products that perfectly fits our e-shop's requirements.

- **Newegg:** While a global standard for hardware, it was ultimately excluded to maintain a more manageable and focused product database.

Scraping Web For Items

Website we will be using for this proyect: https://www.coolmod.com/

To execute the scraping process, we will first need a few python libraries.

1. **Requests:** To send HTTP requests and get the HTML contents from the website
2. **BeautifulSoup4:** To parse the HTML and navigate the document we receive from this using CSS selectors
3. **JSON:** To structure and gather all of the data into a consistent format

Targetted Data Points

Based on the site, the piece of code we will create will focus primarily on the following attributes for each product:

1. **Product Name**
2. **Category**
3. **Current Price**
4. **Original Price**
5. **Availability**

The output file that comes from the python code we created will follow the usual JSON format for compatibility.

Virtual Machine Creation

Virtual Machine 1: Database Server (PostgreSQL)

This section details the setup and configuration of the first node, dedicated to hosting the PostgreSQL database.

1. Virtual Hardware Specifications

- **Operating System:** Linux Mint (Cinnamon/XFCE)
- **Virtualization Platform:** VirtualBox / VMware
- **RAM:** 12069 MB
- **Storage:** 25 GB VDI (Dynamically allocated)
- **Network Adapter:** NAT Network / Bridged (Subject to IP assignment)

2. Operating System Installation

Installed using the official Linux Mint ISO:

- **Installation Mode:** Full disk installation (Erase disk and install).
- **System User:** admin-server
- **Password:** 1234
- **Privileges:** admin-server added to the sudo group.

3. Initial System Preparation

```
sudo apt update && sudo apt upgrade -y
sudo apt install net-tools -y
```

4. Database Engine Installation (PostgreSQL)

```
# Install PostgreSQL
sudo apt install postgresql postgresql-contrib -y

# Verify Service Status
sudo systemctl status postgresql
```

5. Current Progress

- ☑ VM Created with specified hardware.
- ☑ Linux Mint OS installed successfully.
- ☑ User admin-server configured.

- [x] PostgreSQL service is active and running.

Virtual Machine 2: Web Server

1. Virtual Hardware Specifications

- **Operating System:** Linux Mint (Cinnamon/XFCE)
- **Virtualization Platform:** VirtualBox / VMware
- **RAM:** 13306 MB
- **Storage:** 25 GB VDI (Dynamically allocated)
- **Network Adapter:** NAT Network (Matches Database VM for connectivity)

2. Operating System Installation

Consistency maintained with the first node:

- **Installation Mode:** Full disk installation.
- **System User:** admin-server
- **Password:** 1234
- **Hostname:** web-server
- **Privileges:** Full sudo access.

3. Initial System Preparation

```
# Update and upgrade system packages
sudo apt update && sudo apt upgrade -y

# Install network discovery tools
sudo apt install net-tools -y
```

Network Infrastructure Setup

Initially, VMs were using default NAT mode, causing IP conflicts. A dedicated virtual network was created to allow inter-VM communication.

Steps Taken:

1. **Created NAT Network:** Established a global "NatNetwork" in VirtualBox settings with DHCP enabled.
2. **Adapter Configuration:** Switched both VMs' network adapters from "NAT" to "NAT Network".
3. **MAC Unification:** Regenerated the MAC address for the Web Server VM to ensure the DHCP server assigns a unique IP.
4. **Verification:** Confirmed unique IP assignment via `hostname -I`.

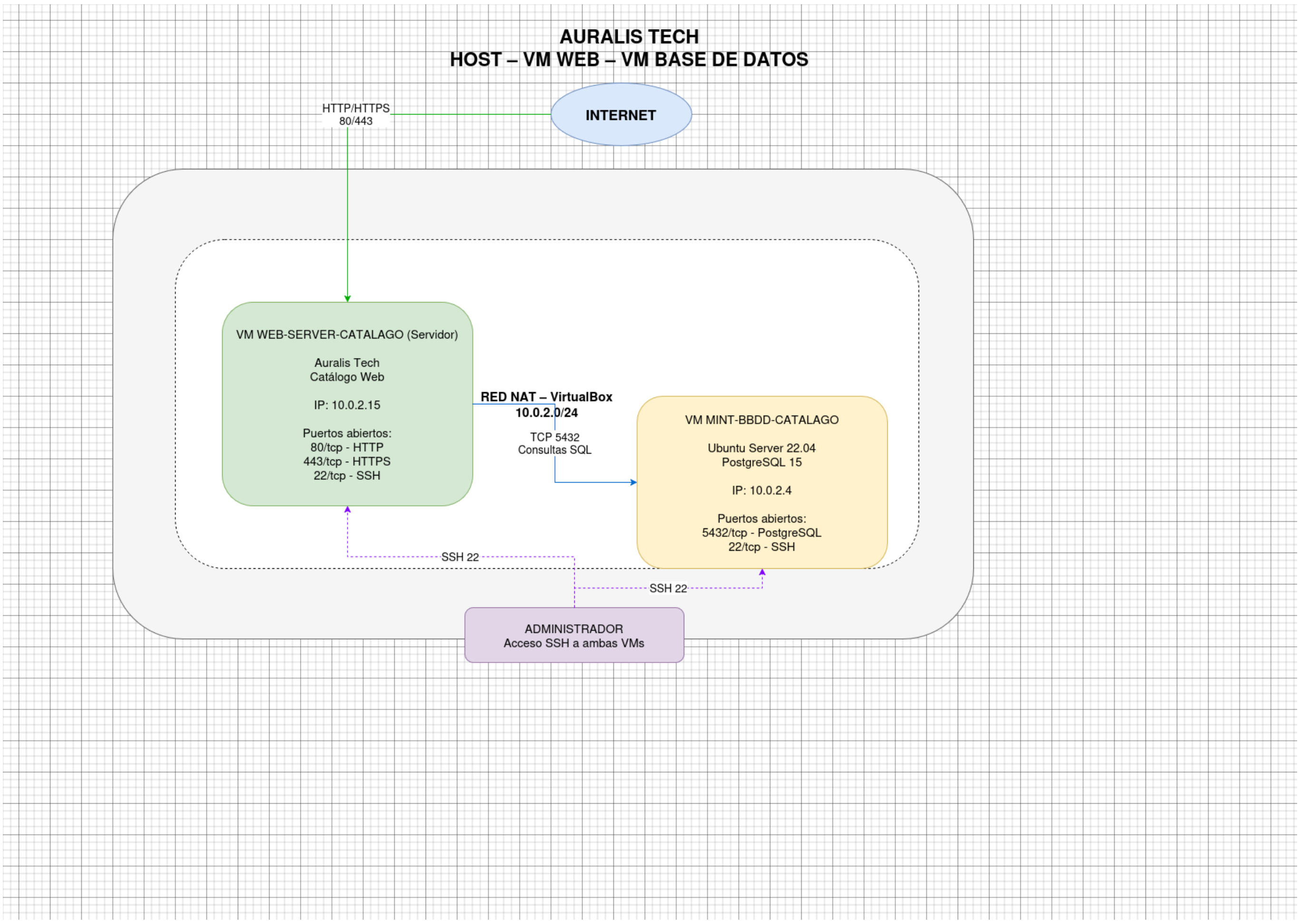
Connectivity Confirmed

The "Destination Host Unreachable" issue was resolved by ensuring both virtual machines were correctly bridged to the same NAT Network.

Final Verification:

- **Source VM (Web):** Successfully reached the Database VM.
- **Network Status:** 0% packet loss.
- **Environment:** Ready for secure service deployment (HTTPS and PostgreSQL remote access).

Project Architecture



Link to repository

https://github.com/samuellugosantos/proyecto-final-1daw/tree/51be1e92a112095035c0ae66889a247a77aa4505